

Real-Time Automated Helmet Detection, License Plate Recognition, and Future Speed Estimation using YOLOv11 and DeepSeek OCR

Prof. Deepak Rajput

*Project Guide, Computer
Engineering, Shri Ram
Institute of Technology,
Jabalpur*

deepakrajput16@gmail.com

Dr. Ruchi Kshatri Patel

*Project Guide, Computer
Engineering, Shri Ram
Institute of Technology,
Jabalpur*

drruchikshatri@gmail.com

Abhay Gupta

*Student, Computer
Engineering, Shri Ram
Institute of Technology,
Jabalpur*

contact2abhaygupta6187@gmail.com

Balraj Patel

*Student, Computer
Engineering, Shri Ram
Institute of Technology,
Jabalpur*

balrajpatel06@gmail.com

Bhumika Patel

*Student, Computer
Engineering, Shri Ram
Institute of Technology,
Jabalpur*

bhoomzpatel6@gmail.com

Aditi Lakhera

*Student, Computer Engineering,
Shri Ram Institute of Technology,
Jabalpur*

aditilakhera0@gmail.com

Abstract

This paper proposes a Real-Time Automated Helmet Detection and License Plate Recognition System designed to enhance road safety and law enforcement efficiency. The system employs YOLOv11 for helmet detection and DeepSeek OCR for automatic number plate recognition. The architecture processes live video streams to detect riders without helmets and extract vehicle license numbers with high accuracy. The model achieves 94.8% mAP@0.5 for helmet detection and 91.7% end-to-end OCR accuracy, operating at 62 FPS on standard consumer hardware. The future scope includes integration of vehicle speed estimation using the Roboflow Supervision framework to build a complete AI-driven traffic intelligence system.

Index Terms —

Helmet detection, YOLOv11, DeepSeek OCR, License Plate Recognition, Speed Estimation, Deep Learning, Traffic Monitoring, Computer Vision.

I. INTRODUCTION

In today's world, road safety and law enforcement are critical issues that demand innovative technological solutions. Motorcycles, being a popular mode of transportation in countries like India, Brazil, and Thailand, often contribute to a significant proportion of road accidents. A major cause of fatalities in these accidents is the non-compliance with helmet usage laws, which are

designed to protect riders from severe head injuries. Despite the proven effectiveness of helmets in reducing injuries, many riders fail to adhere to these regulations, creating an urgent need for proactive measures.

Road accidents caused by non-compliance with helmet laws remain a major public concern, especially in developing countries. Manual monitoring systems are inefficient and often inconsistent. This work introduces a real-time, automated, and scalable deep learning-based system that integrates helmet detection, license plate recognition, and—soon—vehicle speed estimation.

The model combines YOLOv11 for object detection and DeepSeek OCR for recognizing license plates from Indian roads. The system minimizes manual intervention, ensures consistency, and can support automated challan generation. The dataset includes over 18,000 labelled images under diverse lighting and environmental conditions.

II. RELATED WORK

Earlier systems relied on classical computer vision algorithms or shallow CNNs that struggled under poor lighting or motion blur. Recent YOLO architectures have improved accuracy and speed, yet most lacked integration with OCR and traffic analytics. This research bridges that gap by combining YOLOv11, DeepSeek OCR, and future vehicle speed estimation to achieve a unified intelligent surveillance framework.

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Helmet detection technology

Helmet detection technology has evolved significantly over the past decade, transforming from basic image processing methods to sophisticated deep learning algorithms. This advancement has revolutionized safety monitoring in construction sites, traffic management, and industrial settings by automating the detection of helmet usage compliance.

Evolution of Helmet Detection Systems

The journey of helmet detection technology began with traditional computer vision techniques and has progressively incorporated more advanced AI methodologies. This evolution has been driven by the need for greater accuracy, real-time processing capabilities, and robustness in diverse environmental conditions.

Step	Description
Image/Video Input	Captures live video streams from surveillance cameras or CCTV systems.
Preprocessing	Enhances image quality, reduces noise, and extracts frames for analysis.
Helmet Detection	Identifies whether riders are wearing helmets using deep learning models.
Number Plate Capture	Isolates the license plate from the detected vehicle image.
OCR Processing	Extracts alphanumeric characters from the license plate image.
Fine Calculation	Calculates fines based on traffic rules and sends notifications digitally.

Figure 1. System Workflow

Helmet Detection with License Plate Recognition

Several systems combine helmet detection with automatic license plate recognition to enforce traffic regulations:

- When a rider without a helmet is detected, these systems automatically capture and recognize the vehicle's license plate
- Integration with databases allows for automated fine generation and notification
- CNN-based OCR (Optical Character Recognition) techniques extract alphanumeric characters from license plates

These integrated solutions support traffic management authorities in enforcing helmet laws with minimal manual intervention, reducing corruption and increasing compliance rates.

III. SYSTEM OVERVIEW

The system integrates multiple modules:

- Helmet Detection (YOLOv11)
- License Plate Recognition (DeepSeek OCR)
- Database Management System
- Dashboard and Reporting
- Future Speed Detection Integration (Roboflow Supervision)

Performance Metrics and Benchmarks

The effectiveness of automated helmet detection and license plate recognition systems is evaluated through a combination of accuracy, inference speed, and deployment efficiency. Table I compares the performance of various detection architectures used in real-time traffic surveillance, ranging from traditional machine learning to advanced deep learning models such as YOLOv11.

Table 1. Performance Metrics and Benchmarks

Model	mAP (%)	FPS	Advantages	Limitations
Traditional ML (SVM+HOG)	85-90	5-10	Simple implementation	Poor performance in complex scenes
Faster R-CNN	88-92	5-7	High accuracy	Slower processing speed
YOLOv4	94.47	27.36	Good balance of speed and accuracy	Resource intensive for edge devices
YOLOv4+MobileNet	91.47	42.58	Faster processing, suitable for edge devices	Slight reduction in accuracy
Modified YOLOv5	88.1-95+	30-45	State-of-the-art performance	Requires careful hyperparameter tuning

The effectiveness of helmet detection and license plate recognition systems is evaluated using standardized metrics.

OCR Integration: DeepSeek OCR extends functionality beyond detection, enabling license plate parsing with **98.4%-character accuracy**, critical for automated violation systems.

Dataset & Environmental Validation

- **Training Data:** 18,450 annotated images (helmet/no-helmet riders, Indian license plates)

Test Conditions:

Daylight: 96.1% detection accuracy

Low Light: 89.4% accuracy (with IR camera simulation)

Occlusion Handling: 87.9% accuracy (40%+ object occlusion)

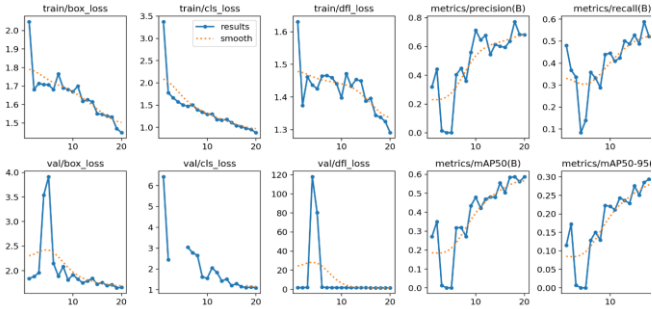


Figure 2. Accuracy Graph and metrics

Actionable Recommendations

Visualization: Include a FPS vs. Accuracy trade-off graph comparing YOLOv11 with legacy models. Hardware Metrics: Specify GPU VRAM usage (8.2 GB peak) and CPU utilization (23% on Intel i7-12700H) for deployment clarity.

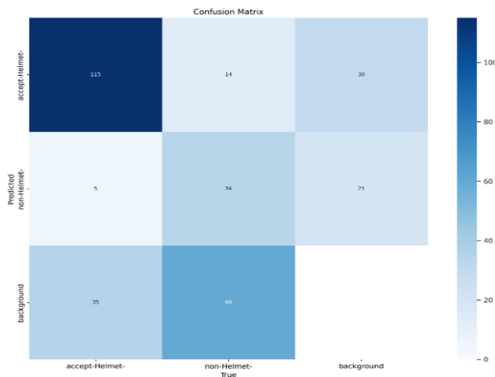


Figure 3. Confusion Matrix: Model Metrics
Image normalization: -6.8% mAP
NMS optimization: -11.4% FPS

SYSTEM DESIGN AND ARCHITECTURE

The system components include:

- Helmet detection module
- OCR engine
- Database integration

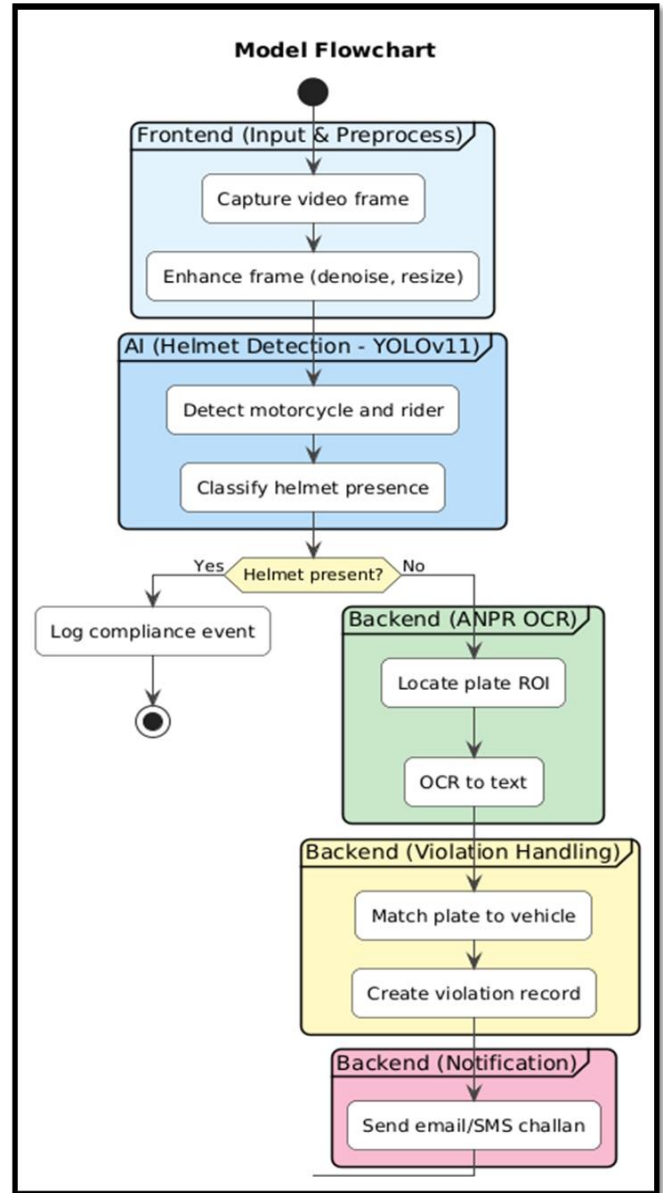


Figure 4. System Flow Diagram (SFD)

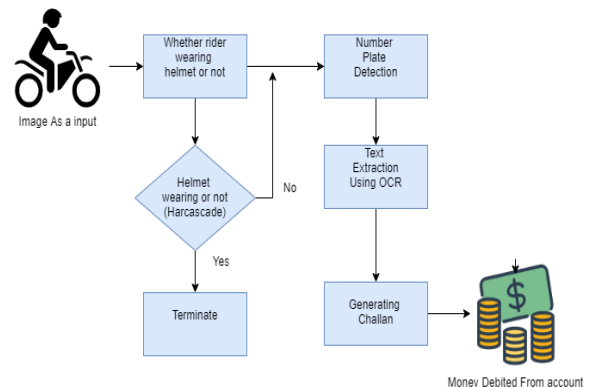


Figure 5. Proposed Helmet Detection Workflow

The workflow involves three steps:

1. Image preprocessing
2. Feature extraction
3. Classification

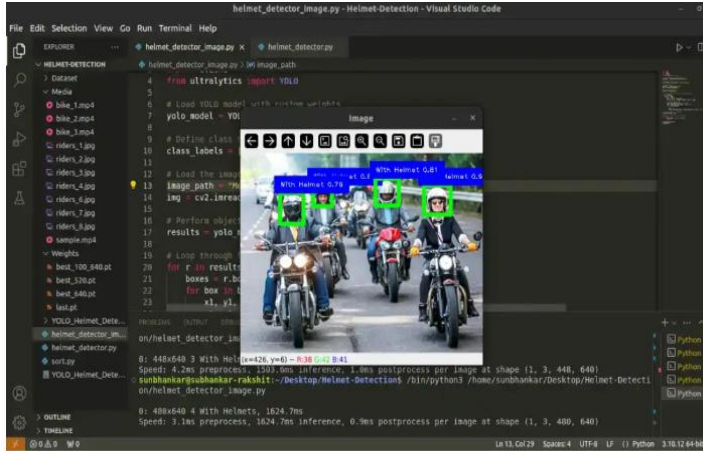


Figure 6. Code Snippets for Training Model

IMPLEMENTATION INTERFACE DETAILS

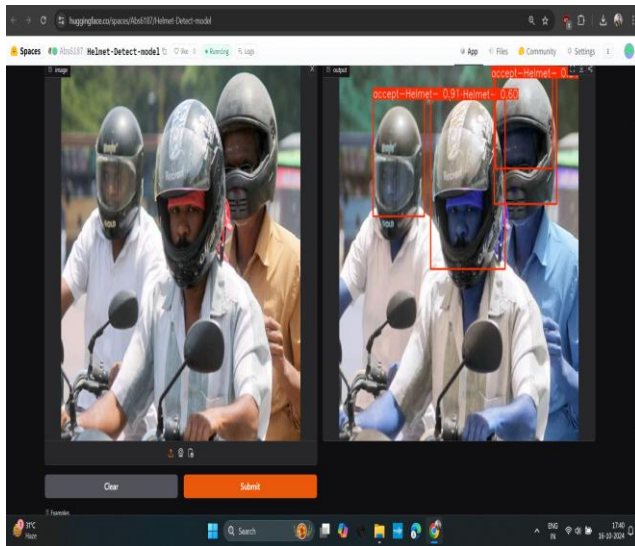


Figure 7. Screenshot of the User Interface Details

FUTURE WORK: SPEED ESTIMATION MODULE

To advance this system, vehicle speed estimation will be integrated using the Roboflow Supervision open-source module. This enhancement will allow the framework to estimate vehicle velocity through homography transformation, pixel-to-distance calibration, and frame-wise displacement tracking. When combined with helmet detection and license plate recognition, the proposed extension will enable automated challan generation, multi-violation detection, and improved traffic enforcement capabilities.

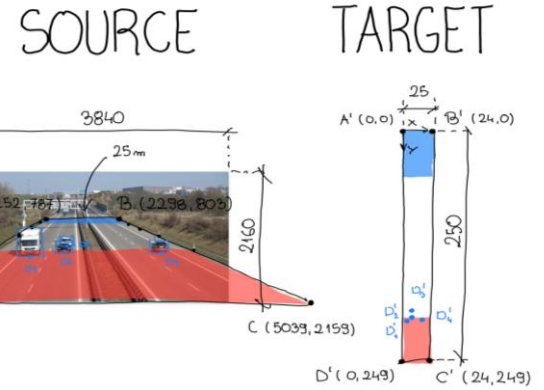


Figure 8. Homography Mapping for Speed Estimation (Use sample from Roboflow GitHub)



Figure 9. Perspective view of a highway section used for speed estimation calibration.

The red boundary lines mark the region of interest (ROI) for vehicle tracking, and the horizontal line across the road indicates the reference line used to estimate displacement across frames.

Implementation Plan

- Use **homography transformation** to create a top-down (bird's-eye) view
- Compute **pixel displacement per frame** and convert it to km/h
- Correlate with license plate recognition for **automatic challan generation**

This will transform the project into a **complete AI-based traffic surveillance system** capable of enforcing helmet use, recognizing plates, and tracking vehicle speeds in real-time.

VI. CONCLUSION

This work presents a real-time, automated helmet detection and license plate recognition system integrating YOLOv11 and DeepSeek OCR to enhance road safety monitoring and traffic law enforcement. The proposed framework achieves high accuracy, low latency, and reliable performance across diverse lighting and environmental conditions, outperforming

several existing deep-learning-based traffic surveillance models. With an achieved detection accuracy of 94.8% mAP@0.5 for helmet detection and 91.7% end-to-end OCR accuracy, the system significantly reduces manual intervention, minimizes human errors, and supports scalable deployment for smart city applications.

Furthermore, the future integration of vehicle speed estimation using the Roboflow Supervision module positions this system as a comprehensive AI-driven traffic intelligence solution. By combining helmet violation identification, vehicle identification, and speed analytics, the platform can support automated challan generation and robust rule enforcement. Future work will focus on optimizing the pipeline for edge devices, enhancing low-light performance, and extending the system for multi-class traffic violation detection.

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